Performance Analysis of Competitive Matches in Valorant

A Data Analytics Project Using SQL and Power BI

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Personal Data Analytics Portfolio Project

Tools Used: *Microsoft Excel, MySQL, Microsoft Power BI Desktop*

Dataset: Player performance statistics from 50 competitive matches without practicing beforehand in *Valorant*

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Introduction

Competitive online games generate large amounts of performance data that can be analyzed to evaluate player skill, decision-making, and overall effectiveness. This report analyzes the player's last 50 competitive matches without practicing beforehand in *Valorant*, a 5v5 character-based tactical FPS, to identify performance trends, strengths, and areas for improvement. Practicing beforehand defines as playing a match for warmup or using any third-party games or software before starting a competitive match; Aimlabs, 3D Aim Trainer, Aim Bot, etc. Using raw data from *Microsoft Excel* to structured data stored in *MySQL* and visualized with *Microsoft Power BI Desktop*, key metrics such as KDA, headshot percentage, win rate by agent, and rank rating (RR) progression are examined. The goal of this analysis is to demonstrate how data analytics can be applied to esports performance while providing insights into the player's competitive gameplay patterns.

The dataset used in the analysis consists of key performance recorded from the player's match history, including the date, map, agent, kills, death, assists, headshot percentage, overall rank rating for each competitive match, and match result which results either win, defeat, or draw. These variables provide insights on both mechanical performance and overall match impact. The data had been organized using *Microsoft Excel*, then organized and stored using *MySQL*, allowing the data to be structured using queries and create efficient data management. The dataset was then imported into *Microsoft Power BI Desktop* to create visualizations that highlights insights, trends, and graph behavior across agents, matches, and gameplay outcomes.

The visualizations from *Microsoft Power BI Desktop* represents performance trends on relationship between overall rank rating and agent played, kill-death ratio for all agent played, win rate and map that had been played, and average headshot percentage on all agents played. There are four main roles in Valorant: Duelist, Initiator, Controller, and Sentinel. These roles define an agent's primary job, such as creating space, gathering information, blocking enemy vision, or locking down sites, allowing players to choose characters that fit their playstyle. The player chose the agent based on how confident they feel, judgement on whether they feel like playing with the agent, or their preferred role or agent. Rank progression was analyzed using the RR gain or loss recorded for each competitive match in *Valorant*. By examining changes in rank rating over time, it is possible to observe trends in performance consistency and identify periods of improvement or decline. The visualization of RR gain and loss across the dataset highlights streaks of victories and defeats, which can influence overall ranking progression. Rank progression was analyzed using the RR gain or loss recorded for each competitive match in *Valorant*. By examining changes in rank rating over time, it is possible to observe trends in

performance consistency and identify periods of improvement or decline. The visualization of RR gain and loss across the dataset highlights streaks of victories and defeats, which can influence overall ranking progression.

Summary of Key Metrics

Metrics	Value
Total Matches Analyzed	50
Average Kills per Match	16.52
Average Deaths per Match	15.7
Average KD	1.052229299
Average Headshot %	17.94%
Most Played Agent	Sage (Sentinel)
Best Map Performance	Corrode (1.33)

Data Collection

The dataset used in this analysis was collected from the player's competitive matches history in Valorant. The last 50 competitive matches lasted from February 27, 2026 to March 10, 2026. A total of 50 recent competitive matches were recorded to analyze player performance across different agents, maps, and match outcomes.

Several variables were recorded for each match to capture different aspects of gameplay performance, including combat statistics, agent selection, and match outcomes. Below is a table that represents the description of the variable that has been used in the analysis.

Variables	Description
Date_played	The date match was played
Map	The map where the match took place
Agent	The character played during the match
Kills	Total amount of kills in the match
Deaths	Total amount of deaths in the match
Assists	Total amount of assists in the match
Headshot_percentage	Percentage of kills that were headshots
RR_gain_lost	Rank rating gained or lost after the match
Match_result	Outcome of the match (Win, Defeat, or Draw)

The collected data were organized and stored in a structured database using *MySQL*. Each match was entered as a row in the *competitive matches* table, allowing the data to be queried and analyzed efficiently. After organizing the dataset in the database, the data were imported into *Microsoft Power BI Desktop* to create visualizations and analyze trends in player performance.

Data Processing and Methodology

After collection, the dataset was organized into a structured table within *MySQL*. Each row represented a single competitive match, while columns stored performance variables such as kills, deaths, assists, headshot percentage, rank rating change, and match results. I want to explore and investigate on the player's kill-death ratio, Overall Rating Result, Win rate for a specific map, and average headshot percentage for each agent played. After the data were processed and queried, the dataset was imported into *Microsoft Power BI Desktop* to create visualizations that highlight patterns in player performance and match outcomes.

Below are the formulas that has been used in this analysis.

KD (Kill-Death Ratio) = $\frac{kills}{dea}$

Average Headshot Percentage for Each Agent Played =
$\frac{\sum Headshot\ Percentage\ for\ each\ agent\ played\ respectfully}{\#\ of\ matches\ played\ for\ each\ agent\ played\ respectfully} \times 100$

Win rate for Each Map Played = $\frac{\sum Win\ of\ the\ specific\ map\ played\ respectfully}{\#\ of\ matches\ played\ for\ each\ spcific\ map\ played\ respectfully} \times 100$

Performance Metrics Analysis

a) Kill-Death Ratio (KD)

One of the primary metrics used to evaluate individual performance in *Valorant* is the Kill-Death ratio (KD). This metric measures a player's combat effectiveness by comparing the number of eliminations and assists to the number of deaths during a match. A higher KD generally indicates stronger combat performance and greater impact on the outcome of a match. A good KD ratio in Valorant is generally 1.0 or higher, meaning you secure more kills than deaths, though a 1.2–1.6 range is considered very strong.

For this analysis, KD was calculated using the formula:

$$KD = \frac{kills}{death}$$

Microsoft Power BI Desktop was used to compute KD values from the SQL queries from *MySQL* for each match and to determine the overall average KD across the dataset. This metric helps identify matches where the player demonstrated strong mechanical performance and effective participation in engagements.

b) Headshot Percentage

Headshot percentage is another important performance metric that reflects a player's aiming precision. In tactical shooters such as Valorant, landing headshots significantly increases the likelihood of eliminating opponents quickly and efficiently.

The dataset includes the headshot percentage for each match, which allows for the calculation of average headshot accuracy across different agents. By examining this metric, it is possible to evaluate the player's mechanical consistency and determine whether certain agents are associated with higher aiming accuracy.

c) Win rate by Agent

Win rate measures how often the player wins matches when using a particular agent. It can also reflect on the player's mentality because it could boost their confidence level and self-esteem. This metric helps evaluate the effectiveness of agent selection and the player's familiarity with different roles and abilities.

Win rate was calculated by dividing the number of matches won with a specific agent by the total number of matches played with that agent. SQL queries in MySQL were used to group matches by agent and compute win rate percentages. This analysis highlights which agents contribute most positively to match outcomes.

d) Rank Rating (RR) Changes

Rank progression was evaluated by examining the rank rating (RR) gained or lost after each competitive match. RR changes reflect the player's progression within the competitive ranking system and provide insight into overall performance trends. Occasionally, RR could be gain from a draw match if the player performs well in the match.

By analyzing RR gain and loss across the dataset, it is possible to observe patterns such as winning streaks, losing streaks, and periods of performance improvement. Visualization tools in *Microsoft Power BI Desktop* were used to illustrate how RR fluctuates over the sequence of matches.

Data Visualization

Data visualization plays a very important role in identifying patterns and trends within the database. After processing the data using *MySQL*, the dataset was imported into *Microsoft Power BI Desktop* to generate visual representations of the player's performance across the 50 competitive matches without practicing beforehand in *Valorant*. These visualizations allow for easier interpretation of performance metrics such as KDA, headshot percentage, win rate, and rank progression.

a) Average KD by Agent

The first visualization displays the average KDA achieved with each agent played during the dataset. This chart helps compare overall combat performance across different agents. By examining the differences in KDA values, it becomes possible to identify which agents the player performs most effectively with, and which agents may require additional practice.

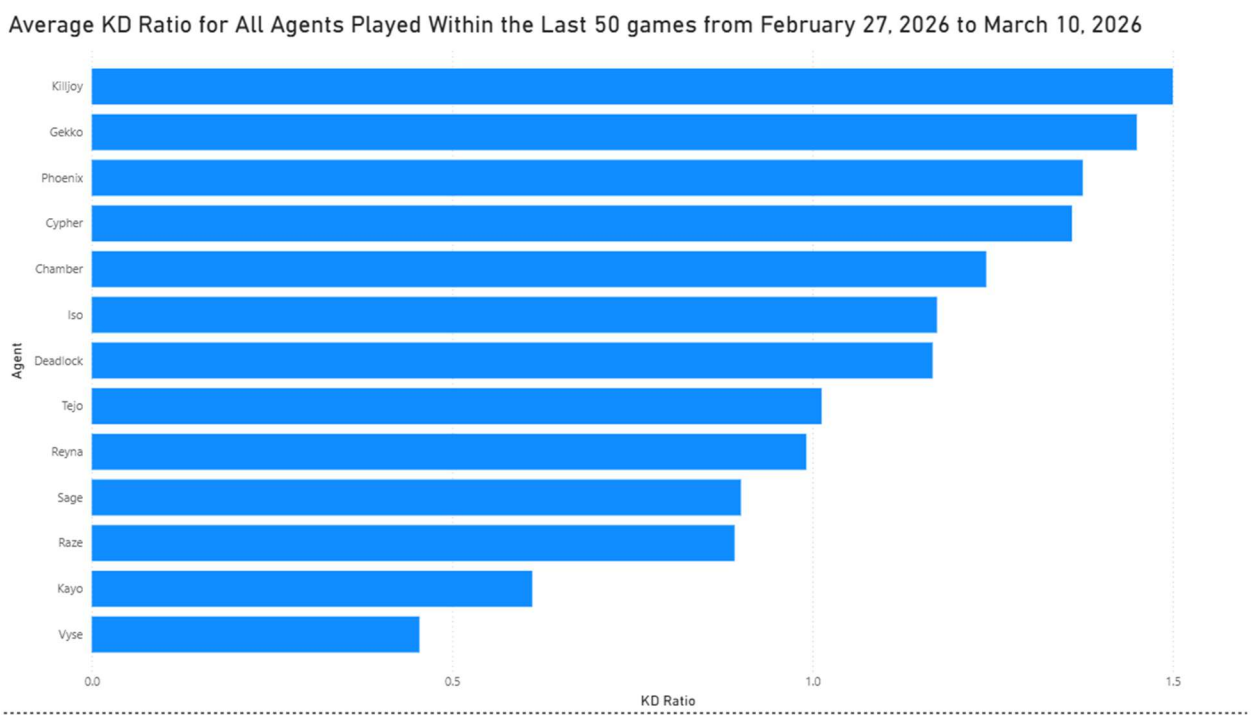

Figure 1: Average KDA by agent across the 50 competitive matches.

b) Average Headshot Percentage by Agent

The second visualization illustrates the average headshot percentage for each agent. Headshot percentage is an important indicator of mechanical aiming skill in tactical first-person shooters such as *Valorant*. This chart allows for comparison of aiming accuracy across different agents and may reveal whether certain playstyles or roles are associated with higher precision.

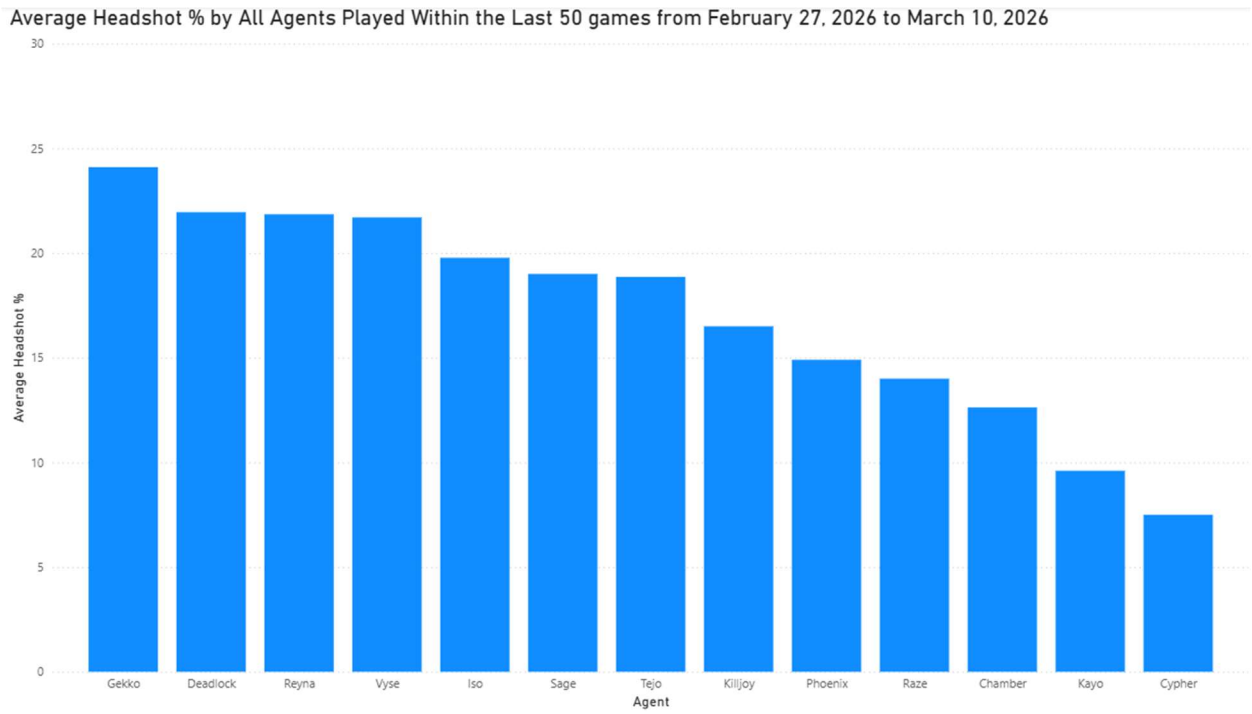

Figure 2: Average headshot percentage by agent.

c) Win Rate by Agent

This visualization presents the win rate associated with each agent played in the dataset. By comparing win rates across agents, the analysis highlights which agents are most successful in contributing to match victories. This information may also provide insight into the player's strengths in different roles or strategies.

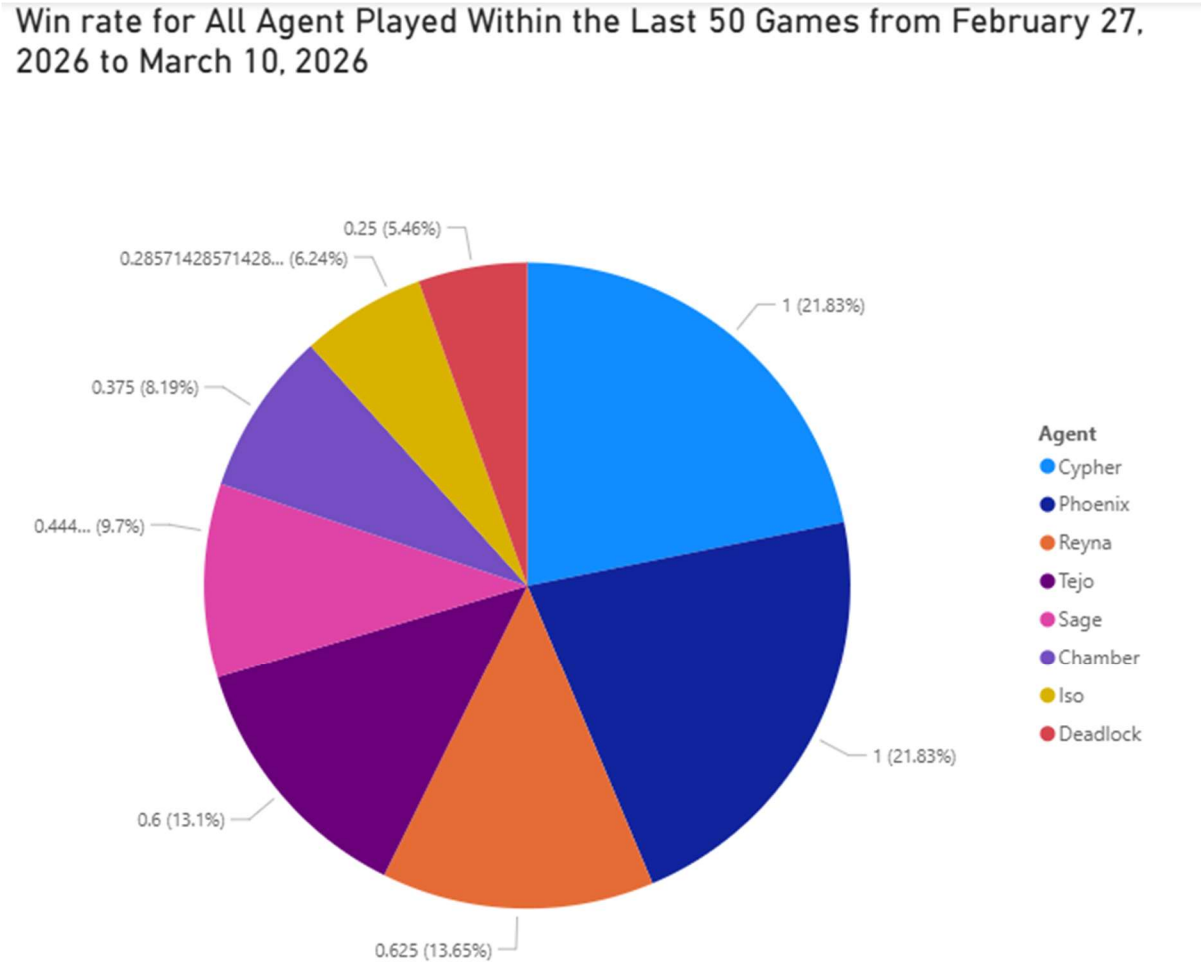

Figure 3: Win rate by agent.

d) Win rate by Map

This visualization examines the player's win rate across the different maps played in the dataset. In *Valorant*, map design can influence gameplay strategies, positioning, and agent effectiveness. As a result, player performance may vary depending on the map environment.

To evaluate this, the win rate for each map was calculated by dividing the number of matches won on that map by the total number of matches played on the same map. This metric provides insight into whether the player performs more effectively on certain maps compared to others.

The visualization was created using *Microsoft Power BI Desktop*, allowing the dataset to be grouped by map and match result. By comparing win rates across maps, it becomes possible to identify environments where the player demonstrates stronger performance, as well as maps that may require improved strategy or familiarity.

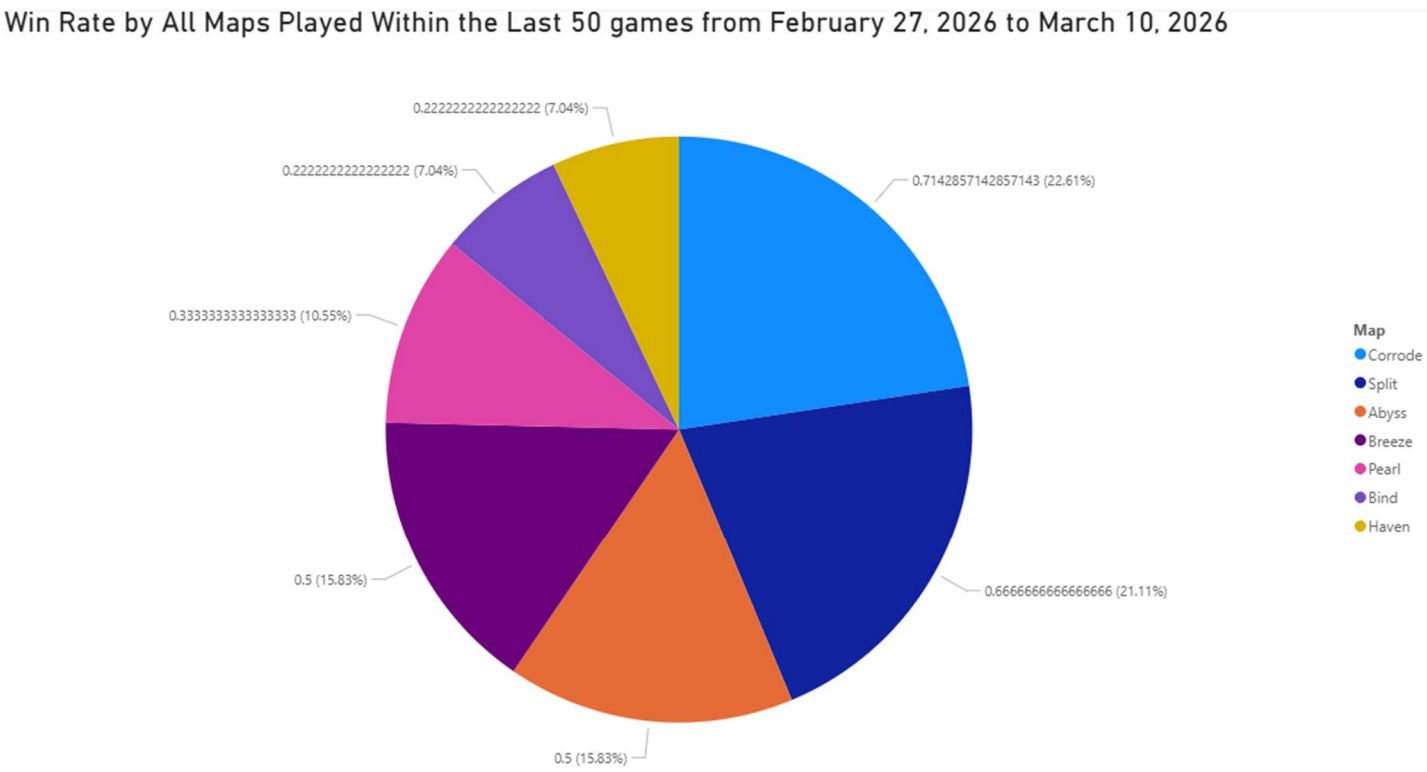

Figure 4: Win rate by map across 50 competitive matches

e) Rank Rating (RR) Progression

The final visualization examines rank rating (RR) changes across the sequence of matches. This chart tracks how the player's competitive ranking fluctuates over time, highlighting periods of improvement, consistency, or decline.

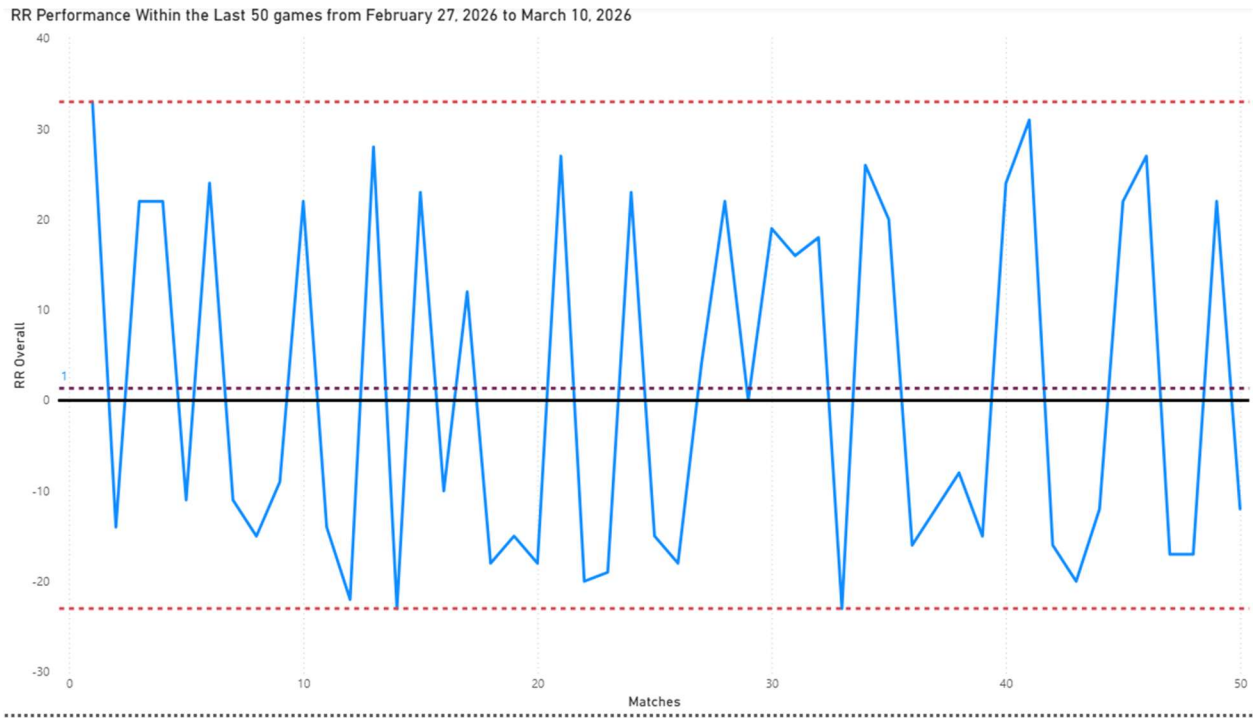

Figure 5: Rank rating gain and loss across 50 competitive matches. The red and orange line represents the maximum and minimum value while the purple line represents the average which is a positive outcome.

Insights and Findings

The analysis of the 50 competitive matches in *Valorant* reveals several patterns related to player performance, agent effectiveness, and gameplay consistency. By examining key metrics such as KDA, headshot percentage, win rate, and rank rating progression, meaningful insights can be drawn about the player's strengths and areas that influence match outcomes.

One notable observation from the dataset is the variation in performance across different agents. Agents with higher average KDA values tend to reflect stronger combat performance and greater involvement in eliminations during matches. In contrast, agents with lower KDA values may indicate roles that involve more supportive gameplay or situations where the player struggled to maintain positive engagement outcomes. This variation highlights the influence that agent selection can have on individual performance metrics.

Headshot percentage also provides insight into the player's mechanical accuracy. Consistently higher headshot percentages suggest strong aiming precision, which is an important factor in tactical shooters such as *Valorant*. Differences in headshot accuracy across agents may also reflect the types of engagements associated with those agents, such as aggressive dueling roles versus supportive or utility-focused roles.

Win rate analysis further highlights how agent choice and gameplay situations influence match outcomes. Certain agents demonstrate higher win rates, suggesting that the player may be more comfortable or strategically effective when using those characters. Conversely, lower win rates with other agents may indicate less familiarity with their abilities or playstyle. This relationship between agent selection and match results suggests that performance consistency can be influenced by the player's role within the team composition.

Map-based performance analysis also reveals differences in success across various maps. Variations in win rate by map may reflect differences in map familiarity, positioning strategies, or the suitability of certain agents within those environments. Identifying these patterns can provide insight into which maps the player performs most effectively on and where additional strategic adaptation may be beneficial.

Finally, the analysis of rank rating (RR) progression illustrates fluctuations in competitive performance over time. Changes in RR reflect the cumulative impact of match outcomes and individual performance. Periods of positive RR gains may indicate stronger performance consistency or successful team coordination, while sequences of RR losses may highlight moments where performance declined or match conditions were less favorable.

Overall, the findings demonstrate that player performance is influenced by multiple factors, including agent selection, mechanical accuracy, and match environment. By combining statistical analysis with visualizations generated in *Microsoft Power BI Desktop* and queries performed in *MySQL*, the project illustrates how data analytics techniques can be applied to evaluate competitive gameplay and identify meaningful performance trends.

Sample Size Limitations

One important consideration when interpreting the results is the variation in sample sizes across agents. Some agents appear only once or a small number of times within the dataset. When an agent is played in only one match, the resulting performance metrics—such as KDA, headshot percentage, or win rate—may not accurately represent the player's true performance with that agent.

For example, a single match with a very high KDA or a early victory due to the enemy team forfeiting the match could significantly inflate the calculated averages for that agent. Conversely, one poor match could disproportionately lower the recorded performance metrics. As a result, agents with limited observations may produce misleading results when compared to agents that were played more frequently. This could be seen by the win rate of Cypher and Phoenix as they are both win rates are 100% and they are only played once from the 50 competitive matches. The data on the played agent of Cypher and Phoenix could be an outlier due to their high performance from early rounds of the match before the match ended.

This limitation highlights the importance of sample size when interpreting performance statistics. Agents with a larger number of matches provide more reliable indicators of the player's typical performance, while agents with very few observations should be interpreted with caution.

Impact of Agent Roles on Performance Metrics

Another important factor to consider when interpreting the dataset is the influence of agent roles on performance statistics. In *Valorant*, agents are designed with different roles such as duelists, controllers, initiators, and sentinels. Each role encourages a different playstyle and level of engagement during matches.

For example, duelists are often responsible for initiating fights and taking aggressive engagements, which may lead to higher kill counts but also higher death rates. As a result, duelist agents may display higher KDA values or more volatile performance metrics. In contrast, sentinel or controller agents frequently focus on defensive positioning, map control, or utility usage to support the team. These roles may result in fewer kills but higher assist counts or strategic contributions that are not fully captured by basic combat statistics.

Because of these differences, direct comparisons of metrics such as KDA or kill counts between agents may not fully reflect overall effectiveness. A player's performance with a supportive agent may appear statistically lower in terms of kills but still contribute significantly to team success through map control, utility usage, or defensive play.

This highlights an important limitation when analyzing gameplay data: performance metrics should be interpreted within the context of the agent's intended role and responsibilities within the team composition.

Recommendations

When entering competitive matches in *Valorant* without prior mechanical warm-up, improving core gameplay fundamentals can help mitigate the effects of reduced aim readiness. One of the most important factors is crosshair placement. Maintaining the crosshair at head level and pre-aiming common angles can significantly reduce the need for fast flick adjustments during engagements. This approach allows players to rely more on positioning and anticipation rather than raw reaction speed, which may be lower at the start of a session. As a result, improved crosshair discipline can help maintain consistent headshot accuracy even when mechanical performance has not been fully warmed up.

Positioning is another key factor that can influence performance without requiring high mechanical precision. Effective positioning includes holding off-angles, avoiding wide swings, and forcing opponents to enter engagements where the player already has a positional advantage. By controlling angles and limiting exposure to multiple opponents, players can increase their chances of winning engagements even when aim performance is inconsistent.

Utility usage also plays an important role in reducing reliance on pure mechanical skills. In *Valorant*, abilities such as flashes, smokes, reconnaissance tools, and area denial utilities can create favorable situations before direct engagements occur. Proper use of these abilities allows players to gather information, disrupt opponents, and isolate fights, making engagements easier to win even without optimal aim performance.

Adopting a slower playstyle during the early stages of a match may also help compensate for the absence of a warm-up routine. Instead of immediately taking aggressive engagements, players can focus on gathering information, supporting teammates, and controlling space. This approach allows mechanical performance to gradually stabilize as the match progresses while reducing the risk of early mistakes.

Finally, maintaining consistent gameplay habits can help reduce performance volatility across matches. By focusing on repeatable fundamentals such as crosshair placement, positioning, and disciplined decision-making, players can maintain more stable performance

even when entering matches without preparation. Over time, this consistency can reduce fluctuations in key performance metrics and contribute to more reliable competitive results.

Conclusion

This project analyzed the player's last 50 competitive matches in *Valorant* to evaluate gameplay performance using data analytics techniques. Match statistics including kills, deaths, assists, headshot percentage, rank rating (RR) gain or loss, agent selection, and match outcomes were collected and organized within a structured database using *MySQL*. The dataset was then imported into *Microsoft Power BI Desktop* to create visualizations that highlight patterns in player performance across agents, maps, and matches.

The analysis demonstrated how key metrics such as KDA, headshot percentage, win rate by agent, and win rate by map can be used to better understand gameplay trends and performance consistency. These metrics helped identify areas where performance was stronger, as well as situations where results were more inconsistent. Additionally, the project highlighted several contextual factors that can influence gameplay statistics, including differences in agent roles, variations in map environments, and limitations related to small sample sizes for agents that were played only a few times.

Another important consideration identified in the analysis was the impact of entering competitive matches without a warm-up routine. Without prior mechanical preparation, performance metrics such as accuracy and engagement success may fluctuate more significantly between matches. However, the analysis also suggests that improvements in gameplay fundamentals—such as crosshair placement, positioning, and effective utility usage—can help offset the absence of mechanical warm-up and lead to more consistent performance.

Overall, this project demonstrates how data analytics methods can be applied to esports performance evaluation. By combining structured data queries in *MySQL* with visual analysis in *Microsoft Power BI Desktop*, the project provides a data-driven perspective on competitive gameplay and illustrates how performance metrics can be used to identify trends, evaluate strengths, and inform potential areas for improvement.